

		force on a small positive test charge.
20	C	By Coulomb's law, electric force between two point charges, Q_1 and Q_2, $F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} = 2.0 \text{ N}$ Electric force between two point charges, Q_3 and Q_2, $F = \frac{q_3 q_2}{4\pi\epsilon_0 r^2}$ $= \frac{2q_1 q_2}{4\pi\epsilon_0 (2r)^2}$ $= \frac{1}{2} \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$ $= \frac{1}{2} \times 2.0$ $= 1.0 \text{ N}$
21	B	Resistance of uniform cylindrical wire